Group Project Phase 2:

Model Building and Evaluation

Group 17

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DECLARATION: *We used chatGPT5.4, claude to format paragraphs, produce drafts, and refine our assignment. We are responsible for the content and quality of the submitted work.*

Problem Definition and Data Preparation

1. Problem Definition and Analytical Approach

The optimal allocation of policing resources has been one of the major challenges, driving the need for predictive models that can accurately identify places with high crime risks. The findings of our previous Exploratory Data Analysis report reveal consistent geographic patterns of crime in Chicago, making it more feasible to predict crime hotspots using historical crime data. To transform the problem into a more actionable task, this project seeks to predict whether a community area will rank in the top 25% for total crime counts across Chicago. This approach not only mitigates the noise in crime data through aggregation by community area but also creates a clear binary classification task that is more effective for our chosen machine learning models.

Regarding the choice of models, given the outperformance of Random Forest (RF) and Gradient Boosting Machines (GBM) identified in the literature review conducted in Phase 1, RF and eXtreme Gradient Boosting (XGB) were selected for model development. This combination leverages the complementary strengths of the two models. XGB is better at making accurate predictions and uncovering non-linear relationships between variables. RF, on the other hand, provides greater resilience against overfitting by averaging the results of several decision trees. Nevertheless, the interpretability of RF and XGB models is limited due to their black-box nature. To address this, a Logistic Regression (LR) model was also introduced. Unlike RF and XGB, which primarily assign importance scores to each feature, LR reveals how each feature contributes to crime risk by returning their coefficients, providing more meaningful insights into crime patterns.

2. Data Preprocessing and Cleaning

Initially, the whole dataset was partitioned into a training dataset (containing 2015–2024 crime data downloaded from the Chicago Crime Dataset) and a testing dataset (containing 2025 crime data). Before being fed into our models, the raw crime dataset was preprocessed, involving removal of irrelevant variables and duplicates, handling missing values, and data type conversions, as well as checking for outliers and invalid values.

2.1 Irrelevant variables and duplicates

The original Chicago crime dataset contains a large number of variables, some of which, such as ID, case number, and IUCR, are irrelevant to the objective of this project and are thus removed. Duplicate crime records can occur due to data entry or system processing errors. To prevent them from distorting statistical analysis and model training, records with identical dates and beat values were considered potential duplicates and then removed.

2.2 Missing values

To ensure the accuracy and reliability of the analysis, multiple strategies were applied to handle missing data. Entries without valid geographic coordinates were excluded since they

cannot be accurately mapped or used in spatial modeling. For location_description, which documents the type of crime scenes, missing data were filled with 'unknown.' As for community area, the missing values were relatively limited, so they were imputed using the mode of the variable to maintain dataset completeness while minimizing potential bias.

2.3 Data Type Conversion

The date column was converted into the datetime64 format, enabling temporal feature extraction. In addition, primary_type (reflecting crime type) and location_description were converted into categorical data types, while community_area and district were converted into integers. These conversions improve computational efficiency and ensure that the dataset can be correctly processed by machine learning algorithms.

2.4 Outlier and invalid value checking

To maintain spatial accuracy, the dataset was examined for abnormal or invalid geographic coordinates. Records with latitude or longitude values outside Chicago's geographic range were removed.

3. Feature Engineering

3.1 Location category aggregation

The original location_description column contains various detailed location types such as street, apartment, and hotel/motel. As discussed in the previous Exploratory Data Analysis report, these fine-grained categories introduce high cardinality that may limit the predictive power of the location_description variable in machine learning models. Hence, location descriptions were grouped into five broader semantic categories, including public outdoor, residential, commercial, institutional, and other. This aggregation reduces noise while preserving meaningful spatial context.

3.2 Consolidation

To ensure the dataset is suitable for the binary classification task, individual crime records were consolidated by community area and year. This process was accomplished through the following procedures. Firstly, in preparation for the 3-year rolling window strategy to be discussed in **section 3.3**, total crime counts over every 3-year period were computed, together with the average latitude and longitude values representing the central location of criminal activities for each community area. The district column was also filled with the mode within each community area. Moreover, the location categories refined in **section 3.1** were converted into ratios, reflecting the proportion of crime incidents occurring in each location category within a community area. Finally, the ratios for the top five most frequent crime types, including theft, battery, criminal damage, assault, and deceptive practice, were calculated with all other crime types grouped into 'other_crime_ratio' to minimize noise from minority crime types.

3.3 The 3-year rolling window strategy

The training dataset spans over the last ten years, making it essential to capture the evolution of crime patterns over time to improve model generalizability. Therefore, a stable 3-year

rolling window approach was adopted. For each target year from 2018 to 2024, features were derived from the preceding three years of crime data. After that, a binary label was created for each community area. Community areas with the top 25% of total crime volume within the target year were labeled as a hotspot (with a value of 1), while all others were labeled 0. The significantly expanded training sample size, due to the implementation of the 3-year rolling window strategy, allowed RF and XGB to learn more complicated crime patterns while enabling LR to obtain more statistically reliable results.

3.4 One-hot encoding and normalization

The district column was transformed into binary dummy variables so that LR treated them as distinct categories instead of numerical values. Apart from that, geographic coordinates including lat_mean and lon_mean, were normalized using MinMaxScaler to ensure the relatively large magnitudes of coordinate data, compared to the location category ratios and crime type ratios, would not distort the models.

Model Implementation & Training

1. Model Architectures

1.1 Logistic Regression

The LR baseline model served as a statistical benchmark for predicting crime hotspots. Despite the data imbalance brought by the top 25% threshold, it was addressed by setting class_weight as "balanced" to prevent the model from attempting to achieve 75% accuracy while sacrificing Recall (the percentage of the actual crime hotspots found by the model).

The model utilized the Limited-memory Broyden–Fletcher–Goldfarb–Shanno solver to find the optimal weights of features by considering how all the variables are related simultaneously. To ensure convergence, the iteration limit was extended to 1,000. Both binary classification results and feature coefficients were generated as the final output with an ROC-AUC of 0.9281, suggesting its excellent performance in distinguishing between hotspots and non-hotspots.

1.2 Random Forest

RF was deployed as a key ensemble model to identify community areas that will rank in the top 25% for total crime incidents occurring across Chicago. Mirroring the baseline LR model, class imbalance was handled by class_weight=balanced. The model performance was optimized when configured with n_estimators=350 and max_depth=10, maintaining a balance between generalizability and complexity of the model. Furthermore, our model utilized parallel processing for training efficiency and set max_features=sqrt to enhance independence between trees. Random Forest was selected for its superior capability to detect non-linear relationships, contributing to high predictive accuracy as demonstrated by an ROC-AUC of 0.9904.

1.3 XGBoost

XGB was implemented as a high-performance gradient boosting model selected specially for its exceptional ability to model sophisticated non-linear patterns through sequential tree

building. The model was tuned using the Bayesian approach with Optuna over 50 trials, and it was ultimately configured with 205 estimators and a maximum depth of 3. The tree structure was relatively shallow, in conjunction with a learning rate of 0.07, aimed to maximize generalizability and avoid overfitting. Similar to the RF model, data imbalance was mitigated by tying the scale_pos_weight to the ratio of non-hotspots to hotspots. Also, by leveraging subsampling and L1 and L2 regularization techniques, the model was forced to learn more universal patterns rather than a few obvious ones, resulting in a high ROC-AUC of 0.9851.

2. Training Process and Hyperparameter Tuning Methodology

2.1 Logistic regression

The LR model was trained with the intention to establish a high-performing linear baseline model. With the aid of the LBFGS solver and implementation of a balanced class weight strategy to mitigate the target imbalance, it was proven to achieve a mean cross-validation ROC-AUC (CV score) of 0.9154 validated through 5-fold Stratified Cross-Validation. In contrast to the RF and XGB models, whose training processes typically require complex hyperparameter tuning methodologies to discover the deep interactions between parameters, LR has fewer parameters and more predictable behaviors. Additionally, since it already achieved a high Recall (0.9000) and testing ROC-AUC (0.9281) with manual configuration, the potential gains from an exhaustive hyperparameter search may be insignificant. Given that, no further tuning was performed on the LR model, allowing it to remain a transparent and interpretable benchmark.

2.2 Random Forest

The training process started with a baseline model with 200 estimators, followed by a two-stage GridSearchCV with a 5-fold Stratified Cross-Validation to further tune the hyperparameters. This methodology optimized model performance and effectively prevented overfitting by testing it against multiple data subsets. By exploring a relatively broad range of values, the first-stage coarse search found the best setting of parameters to be max_depth=10, max_features=sqrt, min_samples_leaf=1, and n_estimators=300. The subsequent fine search narrowed down the parameter space around the preliminary best setting, tested the 48 combinations and consequently identified that the setting with 350 estimators delivered peak performance, increasing the CV score from 0.9701 in coarse search to 0.9713 in fine search.

2.3 XGBoost

Initially, a benchmark model configured with standard guesses was developed. Further, it was refined by leveraging the Optuna-based optimization technique with a 5-fold Stratified Cross-Validation. Compared to the grid search approach employed by RF, Optuna is a more intelligent and efficient method as it prioritizes exploring the combinations of parameters considered more promising. And more importantly, unlike grid search, it searches continuous values that fall between the predefined parameter range instead of being limited to the exact numbers, allowing it to uncover the optimal settings unseen by grid search. Through 50 trials,

the tuning process was finalized with identifying the best setting with 205 estimators and a max depth of 3, which successfully pushed its CV score from the baseline 0.9718 to 0.9750.

Model Evaluation and Comparison

1. Standard Classification Metrics (Appendix A)

While the LR baseline model demonstrates strong performance overall, particularly reflected by high ROC-AUC (0.9281) and Recall (0.9000), its Precision of merely 0.6207 is worrisome. This suggests that roughly 38% of the community areas flagged as hotspots are in fact false alarms, notwithstanding its effectiveness at detecting actual high-crime community areas. Comparatively, the XGB model succeeded in striking a balance between Precision and Recall, resulting in an F1-Score of 0.85. Combined with an Accuracy of 0.9200 and a remarkably high ROC-AUC (0.9851), this model appears to be more reliable than the LR model. But surprisingly, the RF model even outperforms the XGB model. It achieved a nearly perfect ROC-AUC of 0.9904 and an Accuracy of 0.9481, making it the most accurate model among the three. Despite its slightly low Recall (0.8000) relative to the XGB model, it delivers a Precision of 1.0000, indicating that all the community areas labelled as crime hotspots are the actual ones.

Metric	Logistic Regression	XGBoost	Random Forest
Accuracy	0.8312	0.9200	0.9481
Precision	0.6207	0.8500	1.0000
Recall	0.9000	0.8500	0.8000
F1-Score	0.7347	0.8500	0.8889
ROC-AUC	0.9281	0.9851	0.9904

Table 1 Overall Testing Dataset Performance

From the above results, it is evident that no single model dominates in all performance metrics. Therefore, the optimal choice of model largely depends on the specific goal to achieve. For instance, a party with the safety-first perspective may value the LR model most since it can catch 90% of the actual high-risk community areas, reducing the risk of having crime hotspots unidentified. Conversely, the RF model is the ideal choice if the primary goal is to improve resource efficiency. Due to its perfect Precision, the opportunity cost of deploying unnecessary resources to misidentified areas can be eliminated.

1.1 Confusion matrix analysis **(Appendix B)**

To further illustrate the practical consequences of the three models if deployed in the real world, their confusion matrices were analyzed. The results were summarised in **Table 2**. Notably, the LR model wrongly classified 11 safe zones as highly risky (False Positives), which would potentially incur a significant waste of policing resources in a real-world scenario. However, its aggressiveness also contributed to the fewest False Negatives, with

only 2 actual hotspots remaining undetected out of 20 in the testing dataset. On the contrary, the RF model was much more conservative, which produced the fewest False Positives (0) but most False Negatives (4) among the three models. Meanwhile, the XGB model found a middle ground between the previous two models, with only 3 False Positives and 3 False Negatives. Compared to the 11 false alarms generated by the LR model, the XGB model can reduce wasted interventions dramatically while still capturing 85% of actual hotspots.

Confusion Metrix	**Logistic Regression**	**XGBoost**	**Random Forest**	**Actual Totals**	
True Positives	18	17	16	20	Actual Positives
False Negatives	2	3	4		
True Negatives	46	54	57	57	Actual Negatives
False Positives	11	3	0		

Table 2 Summary of Confusion Matrices Results

2. Spatial and Temporal Accuracy (Appendix C)

Since the aggregation of original crime entries at the community area level collapsed individual event timestamps, we mainly focused on testing the models' spatial accuracy. By partitioning the testing dataset into four geographic zones (North-East, North-West, South-East, South-West), the spatial accuracy of the RF and XGB models for each quadrant was analyzed to investigate whether their performance metrics remained consistent across regions.

The XGB model achieved a perfect ROC-AUC of 1.00 in all quadrants except for the North-West, which still maintained an excellent ROC-AUC (0.9762). Its performance in the South-West was particularly strong, with all four metrics hitting 1, suggesting its flawless classification within this region. However, its key limitation lies in the North-East where 50% of the total crime incidents occurred. Notwithstanding a perfect Precision, it struggled with a Recall of 0.6667. It means that approximately 33% of the actual hotspots were missed in this quadrant, remarkably dragging down the overall Recall across the entire map. Notably, the ROC-AUC for all quadrants of the RF model exactly mirrored that of the XGB model. And it maintained its perfect Precision and nearly perfect ROC-AUC in every region and delivered a Recall of 1.0000 in the South-West quadrant. Nevertheless, similar to the XGB model, its Recall (0.6667) also hit the floor in the North-West area, suggesting that the crime incidents in this region were possibly driven by other localized predictors that were not included in the two models.

Overall, both models demonstrate stability across regions, but the RF model slightly outperforms due to more consistency in its Precision and Recall compared to the XGB model. This spatial accuracy analysis proves that the success of the models was not simply averaged across the whole city. Instead, they were effectively learning the unique crime characteristics of different community areas.

3. Model Robustness and Cross-Validation

With the purpose to ensure the above results were more than just a lucky split, a 5-Fold Stratified Cross-Validation was integrated in the training process of all three models. A comparison of the mean CV scores with testing ROC-AUC was performed in **Table 3**. The mean CV score of 0.9154, coupled with the testing ROC-AUC of 0.9281 for the LR model confirm the stability of the model across folds, meaning that the model can generalize well to unseen data without signs of overfitting. Beyond the 5-Fold Stratified Cross-Validation, the XGB and RF model further enhanced their robustness through Optuna tuning and the two-stage GridSearch. The CV scores of the final tuned model were 0.9750 and 0.9713 respectively, highlighting their consistency across samples. Moreover, their respective ROC-AUC of 0.9851 and 0.9904 for the testing dataset were very close to their CV scores, providing evidence of the generalizability and robustness of the two models.

Metric	Logistic Regression	XGBoost	Random Forest
Mean CV Score (i)	0.9154	0.9750	0.9713
Testing ROC-AUC (ii)	0.9281	0.9851	0.9904
Stability (i-ii)	0.0127	0.0101	0.0191

Table 3 Model Stability

Model Interpretation and Business Insights

1. Feature Importance Analysis (Appendix D)

The RF and XGB models share significant similarities in their top 10 feature importances. The theft ratio exhibits the most predictive power in both models. This dominance is most noticeable in the XGB model, where its importance score exceeds 0.16, twice that of the second-ranked commercial ratio. The commercial ratio is also ranked highly in the RF model with similar importance scores. The high ranking of both theft and commercial ratios across the two models indicates that commercial areas with frequent theft are primary indicators of high crime risk. These areas tend to be concentrated with suitable targets lacking alertness or guardianship, creating an environment that is also ideal for other crimes such as criminal damage and deceptive practice. This crime pattern is correctly captured by the two models, demonstrating their predictive power. Another noteworthy feature with pronounced influence across the two models is the mean latitude. As found in the previous Exploratory Data Analysis report (Figure 3) in Phase 1, the crime heat map shows a distinct vertical corridor. Moving across latitudes results in crime density changes that are dramatic and predictable, whereas the shifts across longitudes are relatively irregular, providing less predictive value for the two models. This explains why mean latitude is ranked significantly higher than mean longitude.

One major divergence between the three models is how the district variables, especially district 2 and 6 are ranked. The XGB model gives an importance score around 0.06 to both. Furthermore, the LR model treats them as the most influential variables, with district 6 exceeding a coefficient of 2 and district 2 falling below -2. These suggest that the two districts are strong predictors of hotspots. Specifically, the LR model considers district 2 as the safest area while district 6 signals danger. Nevertheless, the RF model finds these two features less meaningful, assigning them importance scores of around 0.02. This divergence is likely caused by the cardinality bias of the RF model, which tends to overvalue variables with numerous unique values. Therefore, the district features, which were engineered as binary values, may be artificially underrated compared to other variables such as crime ratios.

2. Actionable Insights for Law Enforcement

In order to maximize the efficiency of policing resources allocation, law enforcement agencies should tailor their choice of model to the specific objectives to achieve. The public safety policy makers who prioritize safety would favor the LR model for its highest Recall (0.90) to ensure that the vast majority of actual hotspots can be found. However, for agencies operating under strict budgets, the RF model is preferable given its perfect Precision (1.00). If the stakeholders attempt to maintain a balance between identifying the most actual hotspots and minimizing false alarms, the XGB model offers a middle ground, maintaining a high F1-score (0.85) and a 0.92 Accuracy.

Based on the feature importance analysis, law enforcement agencies may consider prioritizing policing interventions like patrols or expanded surveillance camera coverage in places like commercial areas with high theft ratios where key predictors overlap. Also, the resource allocation can be strategically shifted according to the specific spatial patterns detected. For instance, unnecessary services deployed to safe zones like district 2 should be redirected to dangerous areas like district 6.

3. Limitations and Constraints

This project is subject to the following limitations and constraints. Firstly, despite the effectiveness of the top 25% threshold used to define hotspots, it creates a borderline that overshadows community areas that fall just below the threshold, impairing the relevance of the prediction results. Secondly, aggregating individual crime records by community area and year masks temporal patterns, preventing the models learning from hourly or seasonal trends. Finally, while all three models exhibit predictive power within Chicago, their generalizability to other geographic regions outside of the city remained untested. To address these, in Phase 3 we will consider implementing a weighted scoring system instead of a binary classification, including temporal features such as the ratio of night to day crimes, and testing model performance across various regions in the USA with the National Incident-Based Reporting System dataset (NIBRS) to ensure broader geographic generalizability.

GitHub repository link:

https://github.com/LisazhenYANG/IT5006-Group-Project/tree/main

Appendix A

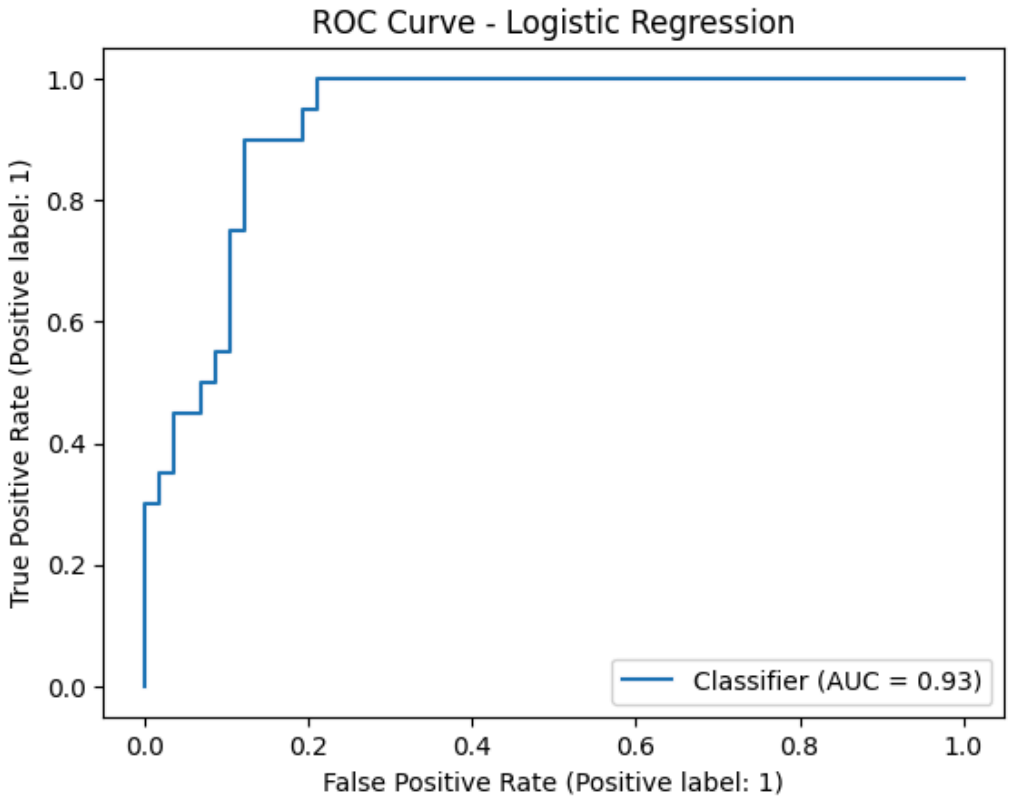

Figure 1 ROC Curve for the Logistic Regression Model

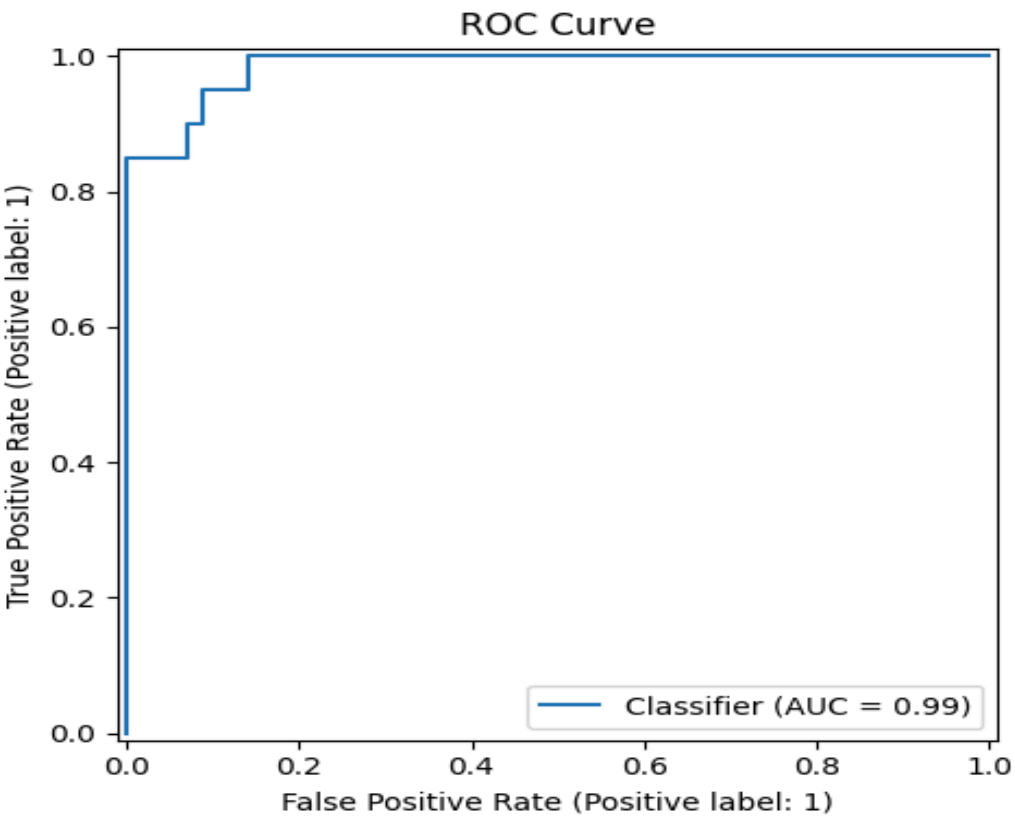

Figure 2 ROC Curve for the XGBoost Model

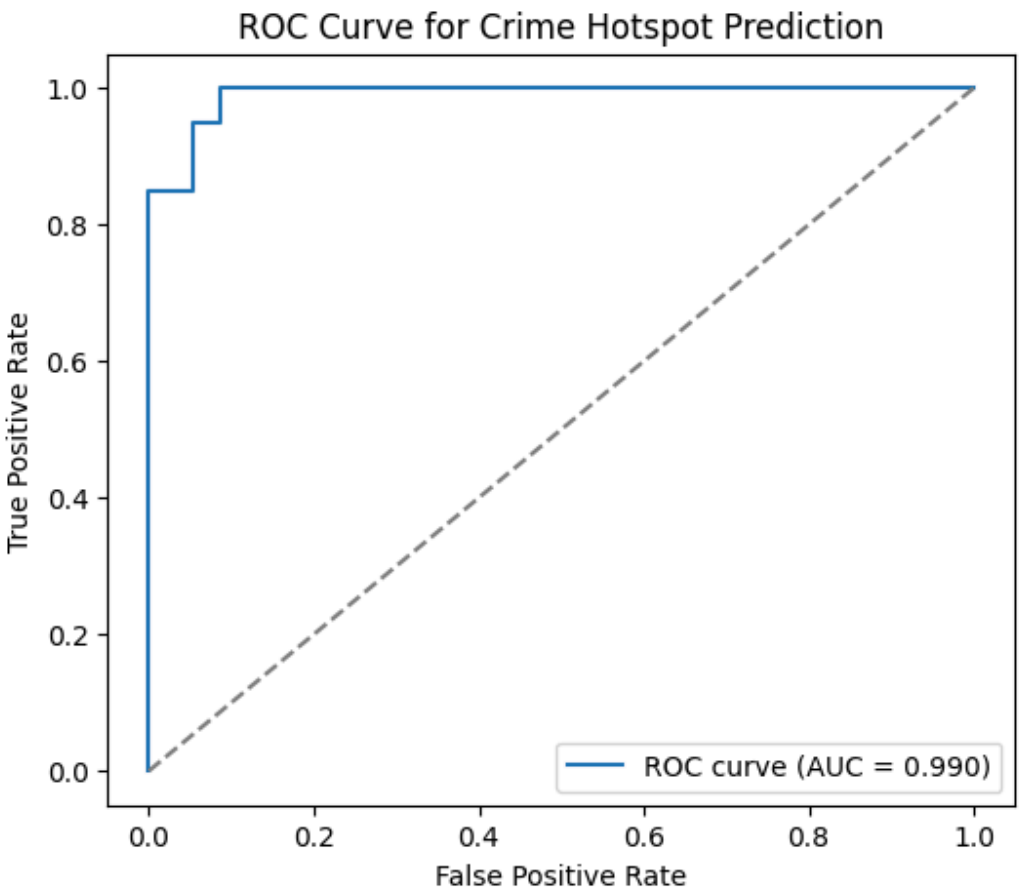

Figure 3 ROC Curve for the Random Forest Model

Appendix B

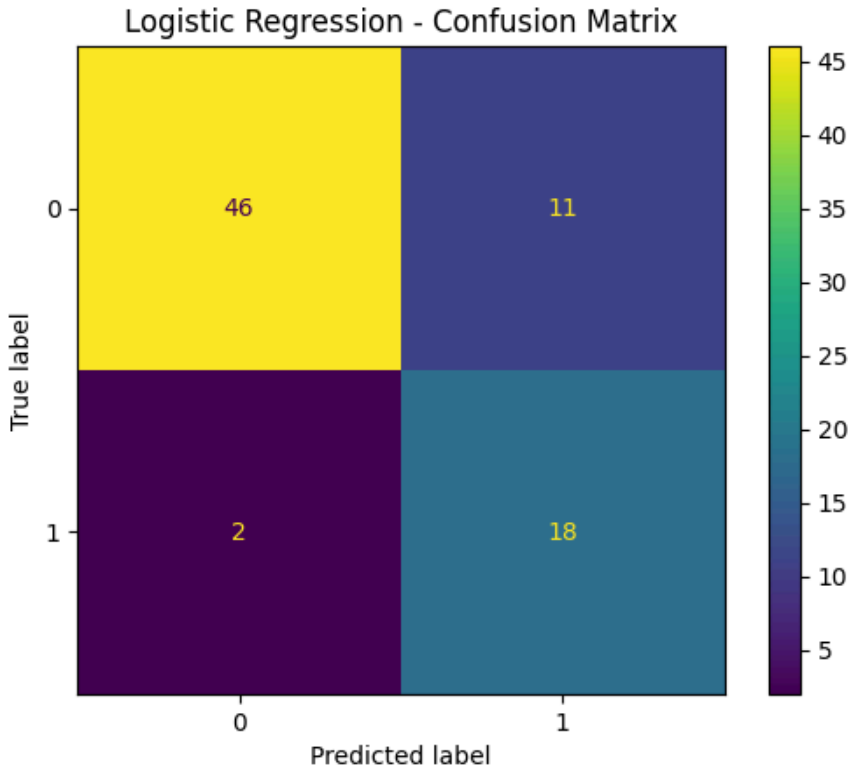

Figure 4 Confusion Matrix for the Logistic Regression Model

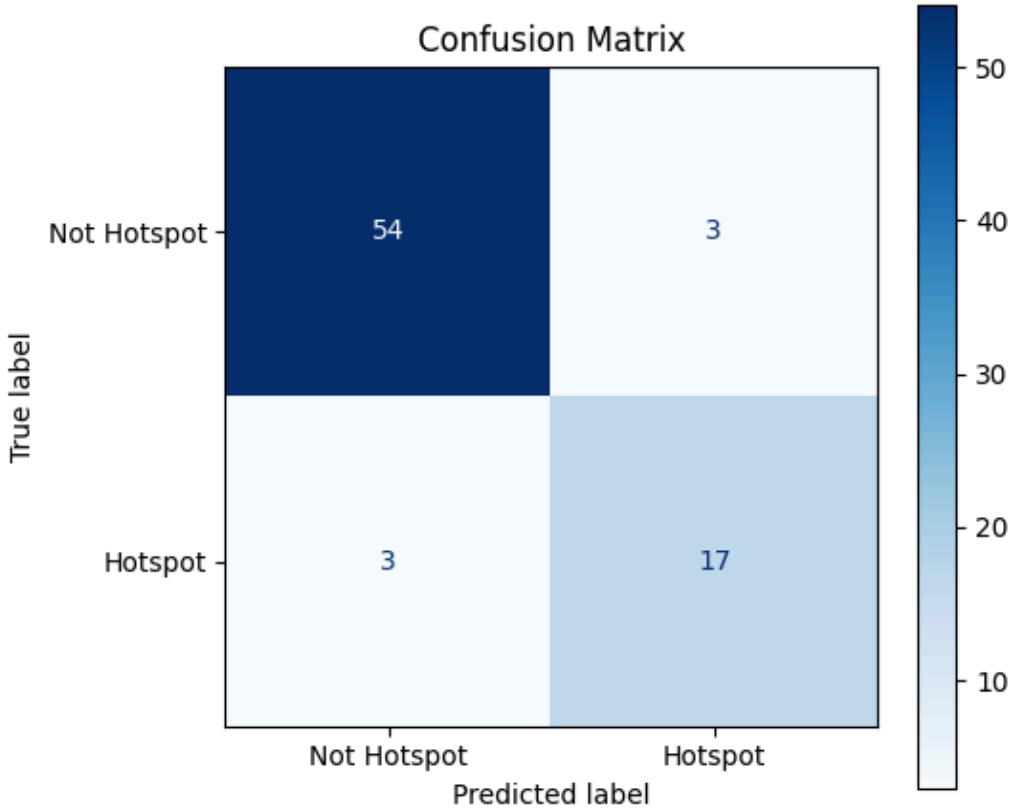

Figure 5 Confusion Matrix for the XGBoost Model

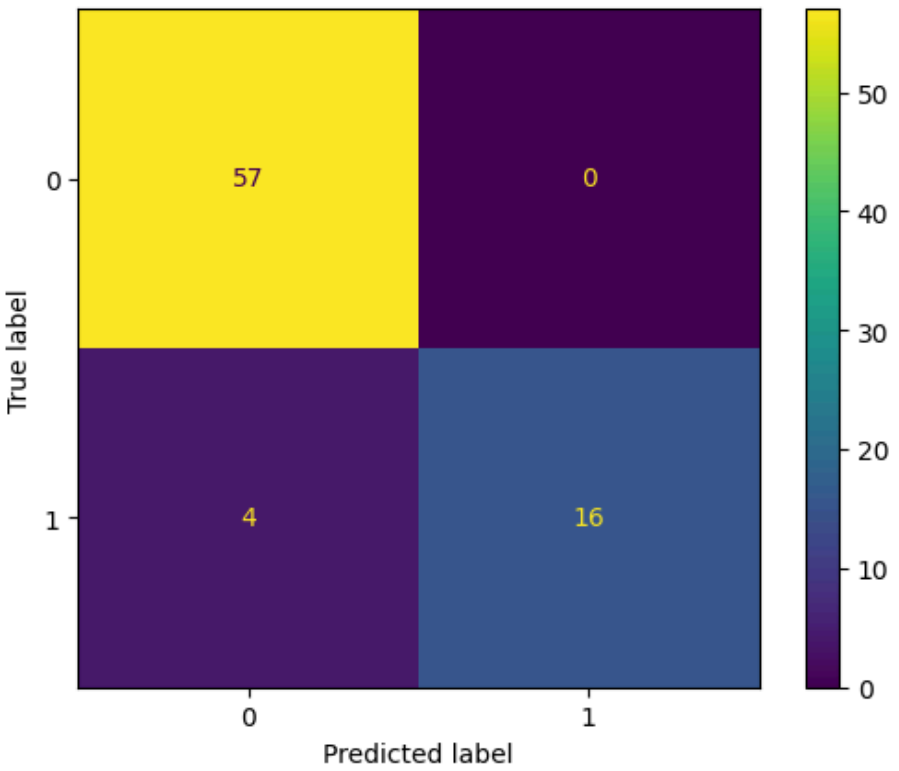

Figure 6 Confusion Matrix for the Random Forest Model

Appendix C

Quadrant	Number of Samples	Percentage of Hotspots(%)	Precision	Recall	F1	AUC-ROC
North-East	12	50.0	1.0000	0.6667	0.8000	1.0000
North-West	27	22.2	0.7143	0.8333	0.7692	0.9762
South-East	27	25.9	0.8750	1.0000	0.9333	1.0000
South-West	11	9.1	1.0000	1.0000	1.0000	1.0000

Table 4 Spatial Accuracy by Quadrant for the XGBoost Model

Quadrant	Number of Samples	Percentage of Hotspots(%)	Precision	Recall	F1	AUC-ROC
North-East	12	50.0	1.0000	0.6667	0.8000	1.0000
North-West	27	22.2	1.0000	0.8333	0.9091	0.9762
South-East	27	25.9	1.0000	0.8571	0.9231	1.0000
South-West	11	9.1	1.0000	1.0000	1.0000	1.0000

Table 5 Spatial Accuracy by Quadrant for the Random Forest Model

Appendix D

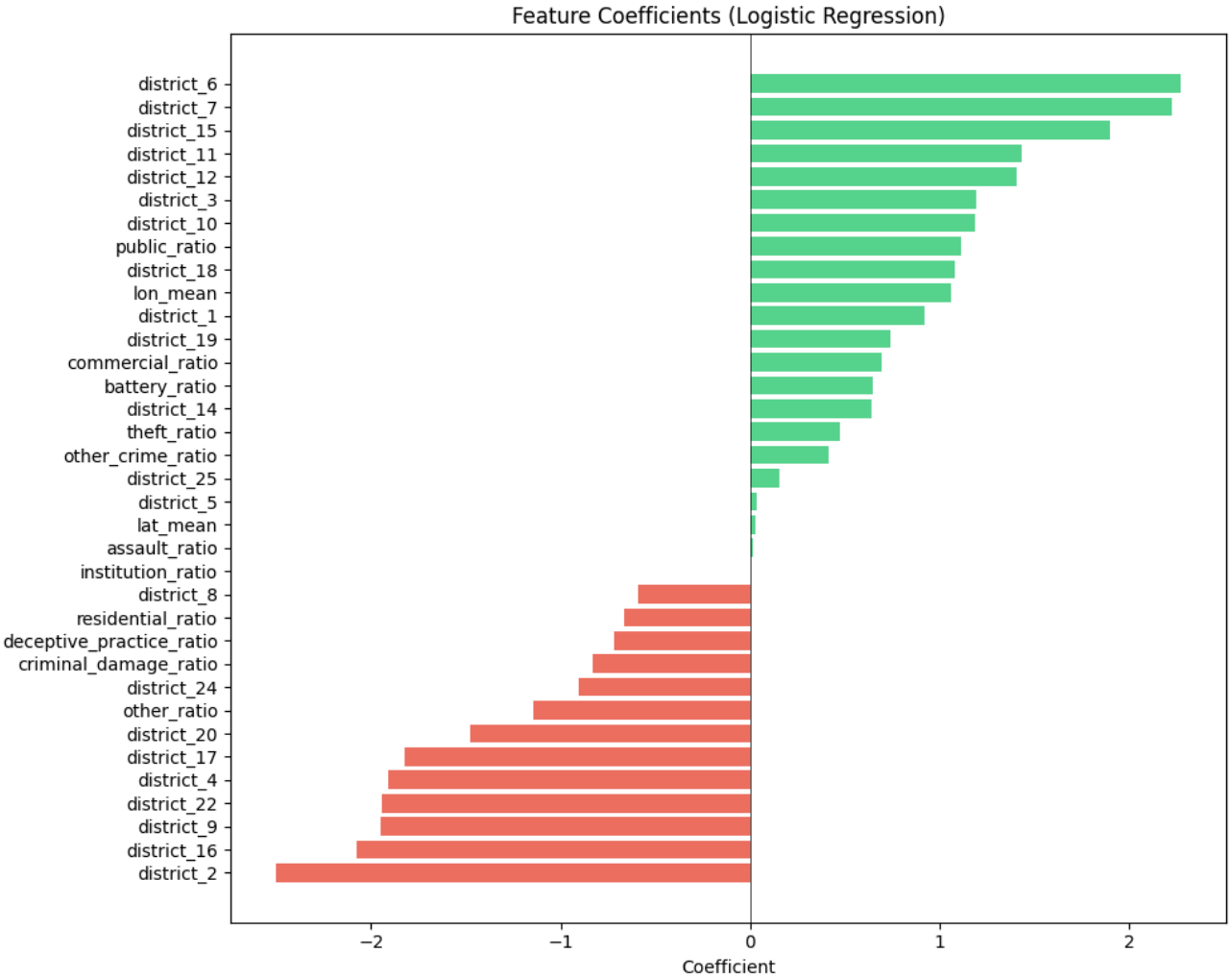

Figure 7 Feature Coefficients for the Logistic Regression Model

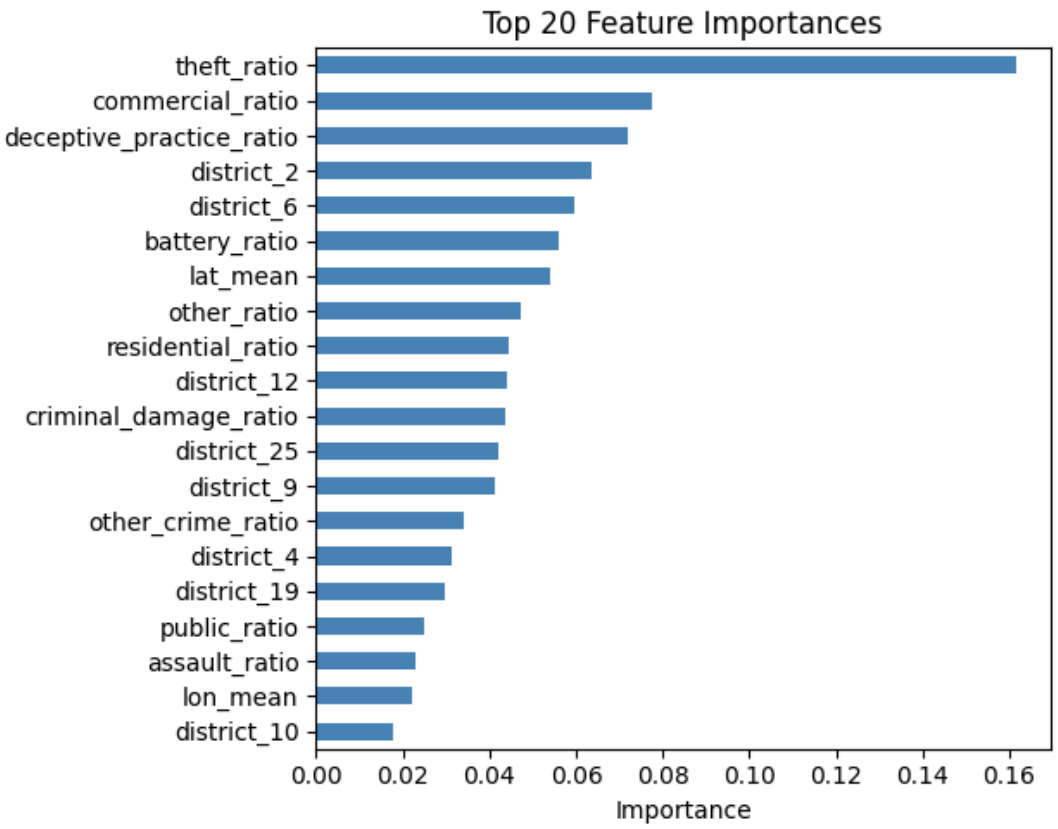

Figure 8 Feature Importances for the XGBoost Model

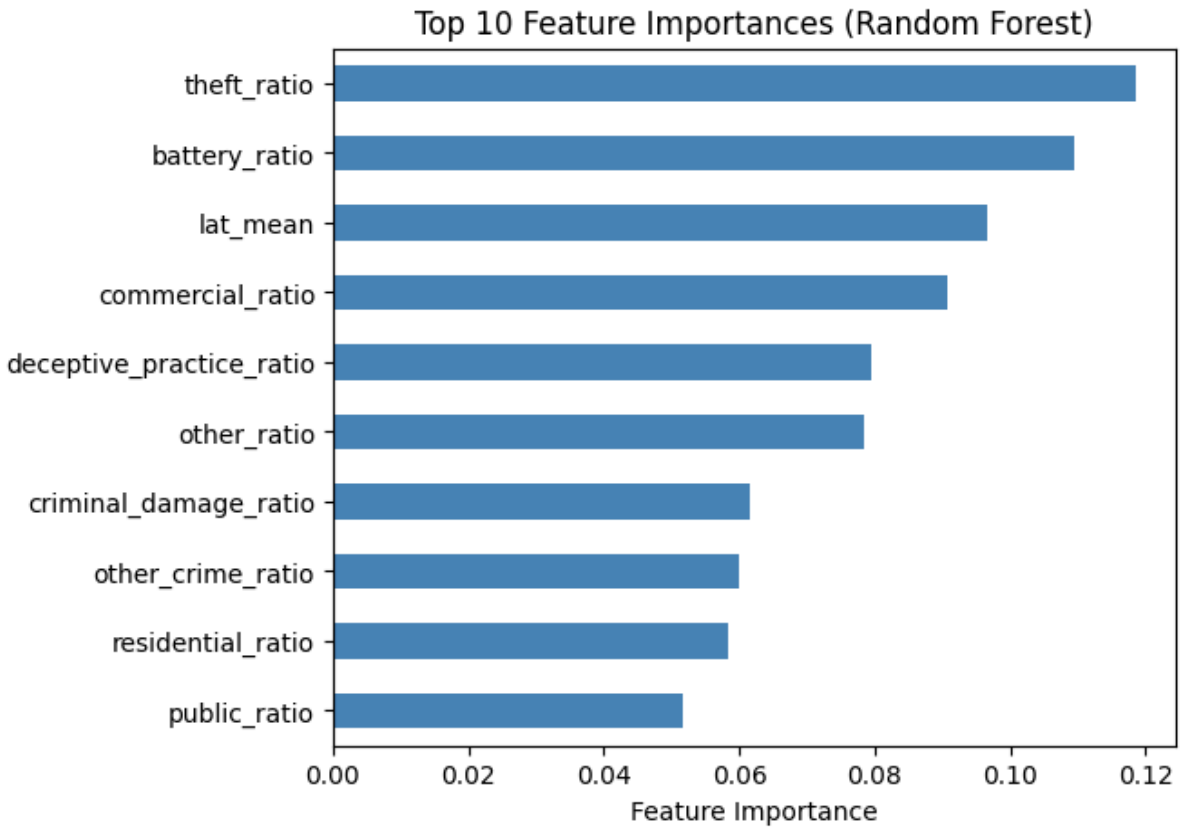

Figure 9 Feature Importances for the Random Forest Model